Rendeiro 2020 CLL — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

4 CLL clones (CLL1/5/6/8, 7 samples) integrate to one continuous mass; sample iLISI 1.0→4.7/7, patient 1.0→2.6/4 at θ=5. Gast→Crest axis reproduces within every patient clone (r=−0.47..−0.59) — shared biology, not a per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

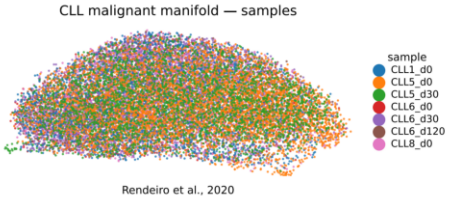
Single Gast→Crest developmental axis: Gast opposes Crest (smoothed r=−0.74), Crest an inverse terminal readout (Crest-high decile pseudotime 0.75 vs Gast 0.29). No Gast/pSom co-expression hub (joint-high enrichment 0.34 < 1). Curve fold-free.

Gate 3 — hub cross-repression → NA

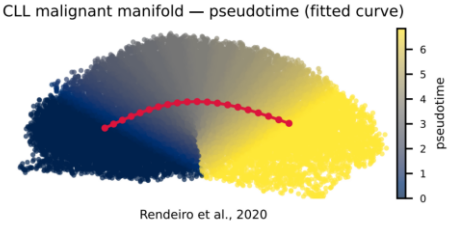
Single collapsed axis, no bifurcation — no two competing arms to test for mutual Gast↔pSom cross-repression.

STEP FIGURES

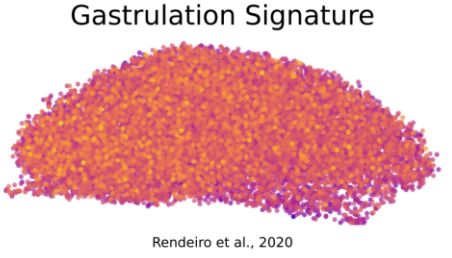
Gate 1: integration



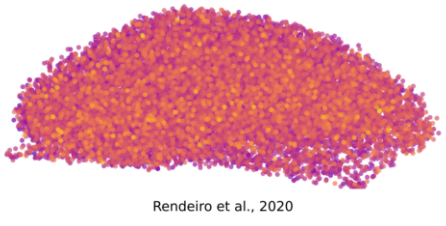
Trajectory / root



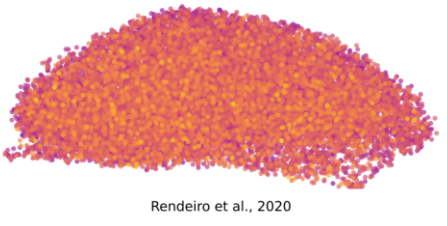
Gast



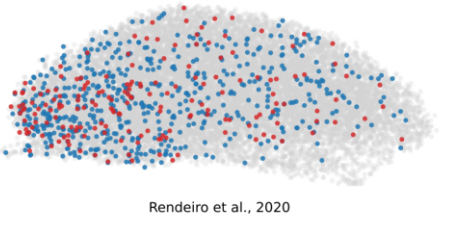
pSom



Crest



EMT-TFs



Dataset. Chronic lymphocytic leukemia (Rendeiro et al., 2020); 10x, longitudinal ibrutinib time-course; 18,854 malignant raw → 18,657 (7 samples ≥100 cells) → 18,490 after island removal (167 detached-speck cells, 2 rounds). 4 patients; 43.8% cycling (axis-orthogonal).

Verdict. ✓ Pattern reproduces as a non-branching embryonic axis: a Gast/pSom-high origin → Crest-high terminus; developmental order Gast ρ=−0.63, Crest terminal (decile 0.75) along pseudotime, confirmed on a diffusion embedding independent of the fitted UMAP (ρ=−0.92). Linear geometry; Gate 3 NA. EMT-TFs near-silent (lymphoid malignancy).

Parameter choices

Min cells/sample	≥100 (gap 1012→80, 12.65x)	**Integration**	Harmony θ=5; PCA20; nn30; spread=1.3
Scope	all 4 patients, one manifold	**Trajectory**	scFates curve N=20 λ=0.3 μ=100 (2 tips/0 forks)
Root node	Gast/pSom-high tip 0	**EMT-TFs**	ZEB2, ZEB1 (both early; only TFs >3% detect)